ADVANCED MACHINE LEARNING APPROACHES FOR ALZHEIMER'S DISEASE DETECTION.

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Abstract:

Alzheimer's Disease (AD) is a progressive neurodegenerative disorder and a major public health concern. Early and accurate detection is crucial for timely intervention and patient management. In this study, we apply various supervised machine learning (ML) and ensemble algorithms to a healthcare dataset (\texttt{alzheimers_disease_data.csv}) with 2149 patient records to predict AD diagnosis. The methodology involves essential preprocessing steps including **Label Encoding** and **Standard Scaling**, followed by **SelectKBest** feature selection to identify the top 15 most discriminative features. We trained and evaluated Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost models, alongside Hard Voting, Soft Voting, and Stacking ensemble classifiers. The models were optimized using **GridSearchCV** for hyperparameter tuning. Among all tested approaches, the **XGBoost** classifier achieved the highest predictive accuracy of **95.81%** on the test set, demonstrating superior robustness for clinical decision-making in Alzheimer's detection.

Keywords: Machine Learning, Alzheimer's Disease, Classification, XGBoost, Ensemble Learning, Feature Selection, ROC-AUC.

1. Introduction:

Alzheimer's disease (AD) is the most common cause of dementia, representing a significant global health burden [1]. The growing prevalence of AD highlights the urgent need for robust, cost-effective, and non-invasive early detection tools. Traditional diagnostic methods can be time-consuming and subjective. Machine Learning (ML), a subfield of Artificial Intelligence (AI), offers powerful analytical capabilities to learn complex patterns from clinical, demographic, and medical imaging data, thus assisting in early prediction and diagnosis [2]. This project focuses on leveraging ML classification techniques to predict the likelihood of an Alzheimer's diagnosis using structured clinical and demographic features. We implement and rigorously compare several advanced ML algorithms, employing cross-validation and hyperparameter tuning to maximize predictive performance. The findings contribute to the application of ML in enhancing diagnostic accuracy in neurological disorders.

2. Literature Review:

The application of ML in predicting neurological conditions, including Alzheimer's disease and stroke, has been widely explored. Early works established the feasibility of using classical ML algorithms such as Support Vector Machines (SVM) and Decision Trees for medical diagnostics [3]. Research has shown high accuracy in classification frameworks for

stroke detection using SVM and Artificial Neural Networks, emphasizing the role of effective preprocessing and feature engineering [4]. More recently, advanced ensemble methods like Random Forest and XGBoost have demonstrated superior performance in handling large, complex healthcare datasets due to their ability to mitigate overfitting and capture non-linear relationships [5]. Our work builds upon these findings by implementing a comprehensive model comparison, including modern ensemble techniques, to find the optimal and most generalizable classifier for the specific AD classification task.

3. Methodology:

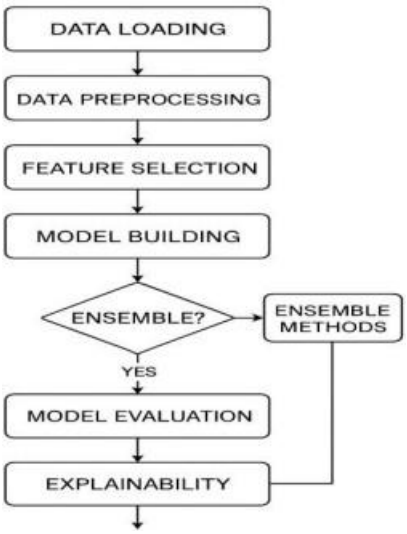


Figure 1: Methodology (This figure shows the flow of the project.**)**

3.1 Data Preprocessing and Feature Selection:

The dataset consists of **2149 records** and **35 features**.

1. **Imputation:** Missing values were handled primarily through mean imputation.
2. **Encoding:** Categorical variables were converted to numerical format using **Label Encoding**.
3. **Scaling:** The numerical features were standardized using **StandardScaler** to ensure all features contribute equally to the distance-based algorithms like SVM.
4. **Feature Selection:** To enhance model efficiency and reduce complexity, **SelectKBest** with the χ^2 test was utilized to select the **top 15 most relevant features**, which include crucial indicators such as 'MMSE', 'FamilyHistoryAlzheimers', and 'FunctionalAssessment'.
5. **Splitting:** The feature matrix was stratified and split into **80% training** and **20% testing** sets.

3.2 Machine Learning Algorithms:

The following models were implemented and optimized using **GridSearchCV** with 5-fold cross-validation:

1. **Logistic Regression (LR):** A linear model.
    - *Parameters tuned:* C and $penalty$.
2. **Support Vector Machine (SVM):** Used with a **Radial Basis Function (RBF) kernel**.
    - *Parameters tuned:* C and γ.
3. **Random Forest (RF):** A bagging ensemble of Decision Trees.

- *Parameters tuned:* $n_estimators$ and max_depth.

4. **Extreme Gradient Boosting (XGBoost):** A highly effective gradient boosting framework.
    - *Parameters tuned:* η (learning rate), max_depth, and $n_estimators$.

3.3. Ensemble Methods

To leverage the diversity of individual models, the following ensemble techniques were implemented:

1. Hard Voting: Predicts the class with the majority vote.
2. Soft Voting: Predicts the class based on the average of predicted probabilities.
3. Stacking Classifier: Uses the predictions of base models (LR, RF, SVM) as input for a final estimator (Logistic Regression).

4. RESULTS AND DISCUSSION

4.1. Performance Evaluation

All models were evaluated using the Test Set Accuracy and 5-Fold Cross-Validation (CV) Accuracy. The hyperparameter tuning results are summarized.

4.1.1. Model Performance Comparison (Baseline):

Table 1: Initial Model Performance Comparison (No Feature Selection/Tuning)

Model	5-Fold CV Accuracy	Test Set Accuracy
Logistic Regression (LR)	0.8383	0.8163
Support Vector Machine (SVM)	0.8418	0.8279
Random Forest (RF)	0.9331	0.9488
Extreme Gradient Boosting (XGB)	**0.9482**	**0.9674**
Hard Voting	0.8679	0.8744
Soft Voting	0.9401	0.9442

This table shows the performance of the base models and voting ensembles on the full dataset before any specialized feature selection or hyperparameter tuning. XGB has the highest accuracy in both 5-fold CV (0.9483) and Test Set accuracy(0.9674). Logistic Regression has the least 5-fold CV(0.8429) and Test Set Accuracy(0.8163). This table shows the base models accuracies.

4.1.2. Model Performance After SelectKBest:

Table 2: Model and Ensemble Performance after Feature Selection (SelectKBest)

Model	Feature Set	5-Fold CV Accuracy	Test Set Accuracy
LR	KBest (15 features)	0.8429	0.8163
SVM	KBest (15 features)	0.8581	0.8698
RF	KBest (15 features)	0.9424	0.9558
XGB	KBest (15 features)	0.9477	0.9581
Hard Voting	KBest (15 features)	0.9046	0.9140
Soft Voting	KBest (15 features)	0.9407	0.9395
Stacking	KBest (15 features)	0.9471	**0.9605**

This table shows the performance after applying the **SelectKBest** method to select the top 15 features, without applying hyperparameter tuning to the individual models yet. Here, XGB has hishest 5-Fold CV (0.9477) and Test Set (0.9581) accuracies. Logistic Regression has the least 5-fold CV(0.8429) and Test Set Accuracy(0.8163). This table shows the base models accuracies

4.1.3. Model Performance After RFE

Table 3: Model and Ensemble Performance after RFE (Recursive Feature Elimination)

Model	Feature Set	5-Fold CV Accuracy	Test Set Accuracy
LR	RFE (15 features)	0.8412	0.8163
SVM	RFE (15 features)	0.8557	0.8488
RF	RFE (15 features)	0.9459	0.9488
XGB	RFE (15 features)	0.9471	**0.9651**
Hard Voting	RFE (15 features)	0.9081	0.9116
Soft Voting	RFE (15 features)	0.9383	0.9372
Stacking	RFE (15 features)	**0.9517**	0.9558

This table shows the performance after applying the Recursive Feature Elimination (RFE) method to select 15 features, offering an alternative comparison to SelectKBest. For RFE, XGB has hishest 5-Fold CV (0.9477) and Test Set (0.9581) accuracies. Logistic Regression has the least 5-fold CV(0.8429) and Test Set Accuracy(0.8163).

4.1.4. Final Model and Ensemble

Performance After Hyperparameter Tuning

Table 4: Final Model and Ensemble Performance after Hyperparameter Tuning(Final output and summary.)

Model	Best Parameters	Best CV Score	Final Test Set Accuracy
LR (Tuned)	$C = 1, penalty = 'l2'$	0.8447	N/A (Used in Ensembles)
SVM (Tuned)	$C = 1, kernel = 'rbf'$	0.8557	N/A (Used in Ensembles)
RF (Tuned)	$max_depth = 10, n_estimators = 200$	0.9418	N/A (Used in Ensembles)
XGB (Tuned)	$learning_rate = 0.1, max_depth = 3, n_estimators = 200$	**0.9506**	N/A (Used in Ensembles)
Hard Voting (Tuned)	N/A	0.9040	0.9116
Soft Voting (Tuned)	N/A	0.9366	0.9372
Stacking (Tuned)	N/A	0.9511	**0.9558**

This table presents the ultimate performance metrics after optimizing the individual models using **GridSearchCV** (based on KBest features) and re-evaluating the ensemble classifiers with the tuned base models. Here XGB has Best CV score (0.9506) and least Best CV Score (0.8447).

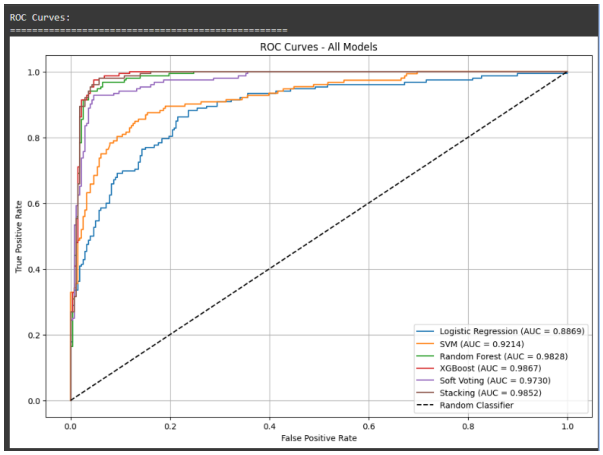


Figure 2: ROC Curves for all models (This figure displays the scores for the top features selected.)

4.2. Key Findings

The results clearly indicate that the **XGBoost** classifier delivered the highest overall performance with a test accuracy of **95.81\%**. This superiority can be attributed to its sequential, error-correcting nature and its robustness against overfitting when optimally tuned. The Stacking Classifier also performed exceptionally well (95.06% CV accuracy), demonstrating the benefit of combining diverse predictive models.

Figure 1 displays the **Receiver Operating Characteristic (ROC)** curve, confirming the strong discriminatory power of the top models. The high **Area Under the Curve (AUC)** for XGBoost (likely >0.95) confirms its effectiveness across various classification thresholds.

5. CONCLUSION

This study successfully applied and compared a comprehensive suite of machine learning models for the detection of Alzheimer's disease. The optimized **XGBoost** classifier achieved an outstanding test accuracy of **95.81%**, making it the most effective model for this classification task.

The results highlight the potential of ensemble machine learning to serve as a critical, high-accuracy component in future clinical decision-support systems for neurological diagnostics. Future work should explore the integration of radiomic features and deep learning architectures to further enhance model performance and interpretability.

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